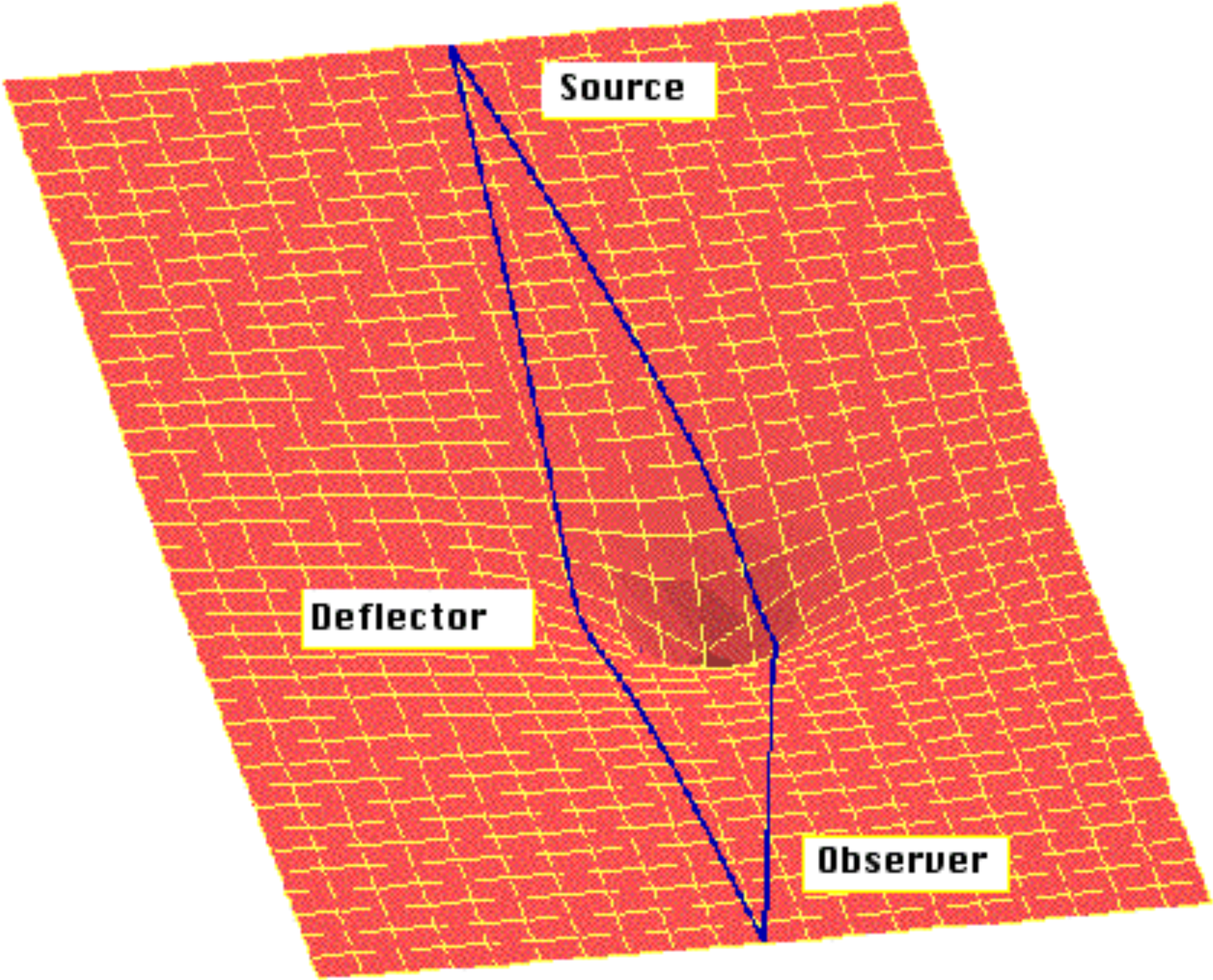
Galaxies and Extragalactic Astronomy

8. Gravitational Lensing

8 Gravitational Lensing

8.1 Introduction

- According to GR matter distorts spacetime and therefore light paths are changed: gravitational lensing
- Gravitational lensing effects depend on the matter distribution and the geometry


Source
Deflector
Observer

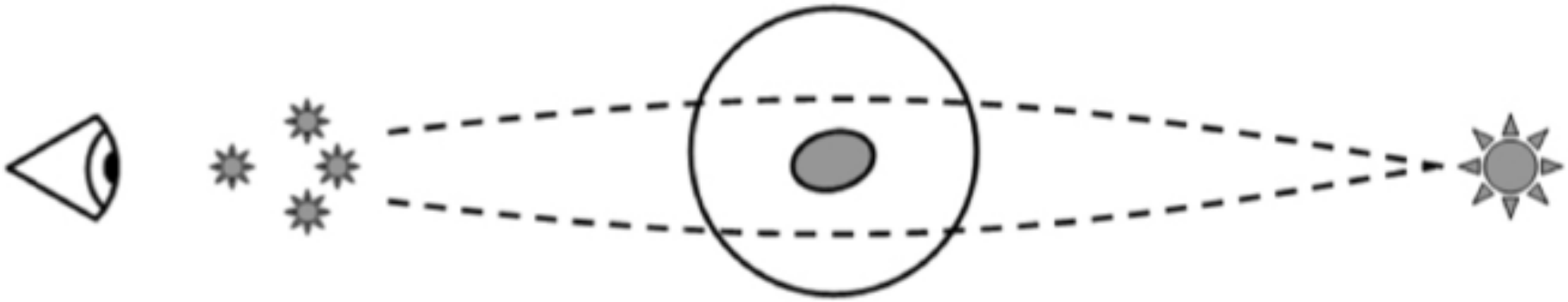
8 Gravitational Lensing

8.1 Introduction

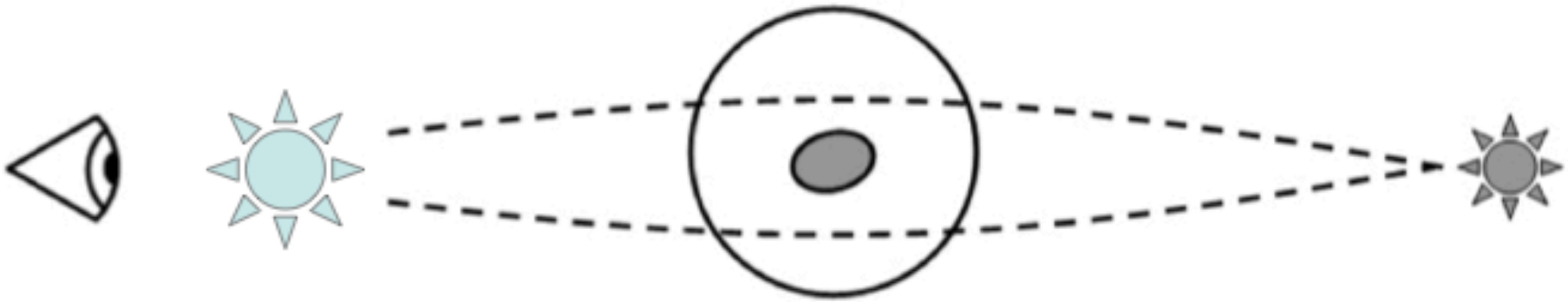
- According to GR matter distorts spacetime and therefore light paths are changed: gravitational lensing
- Gravitational lensing effects depend on the matter distribution and the geometry
- Two regimes: strong and weak lensing
- achromatic

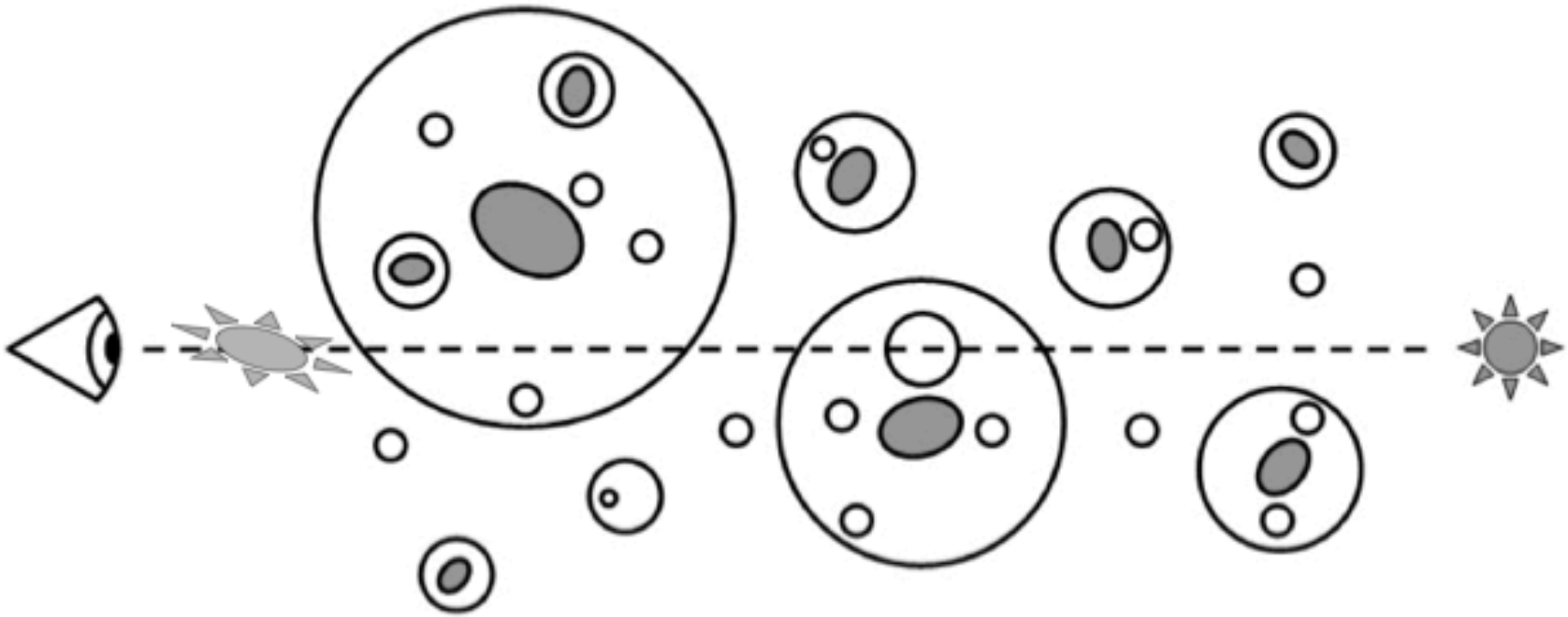
- strong lensing
  - lensing of background sources by foreground galaxies, clusters, …

    (→ strong distortion, magnification, and multiple images)



- microlensing
  - mainly referred to as lensing by objects of stellar (point) masses (→ no distortion, mainly magnification)

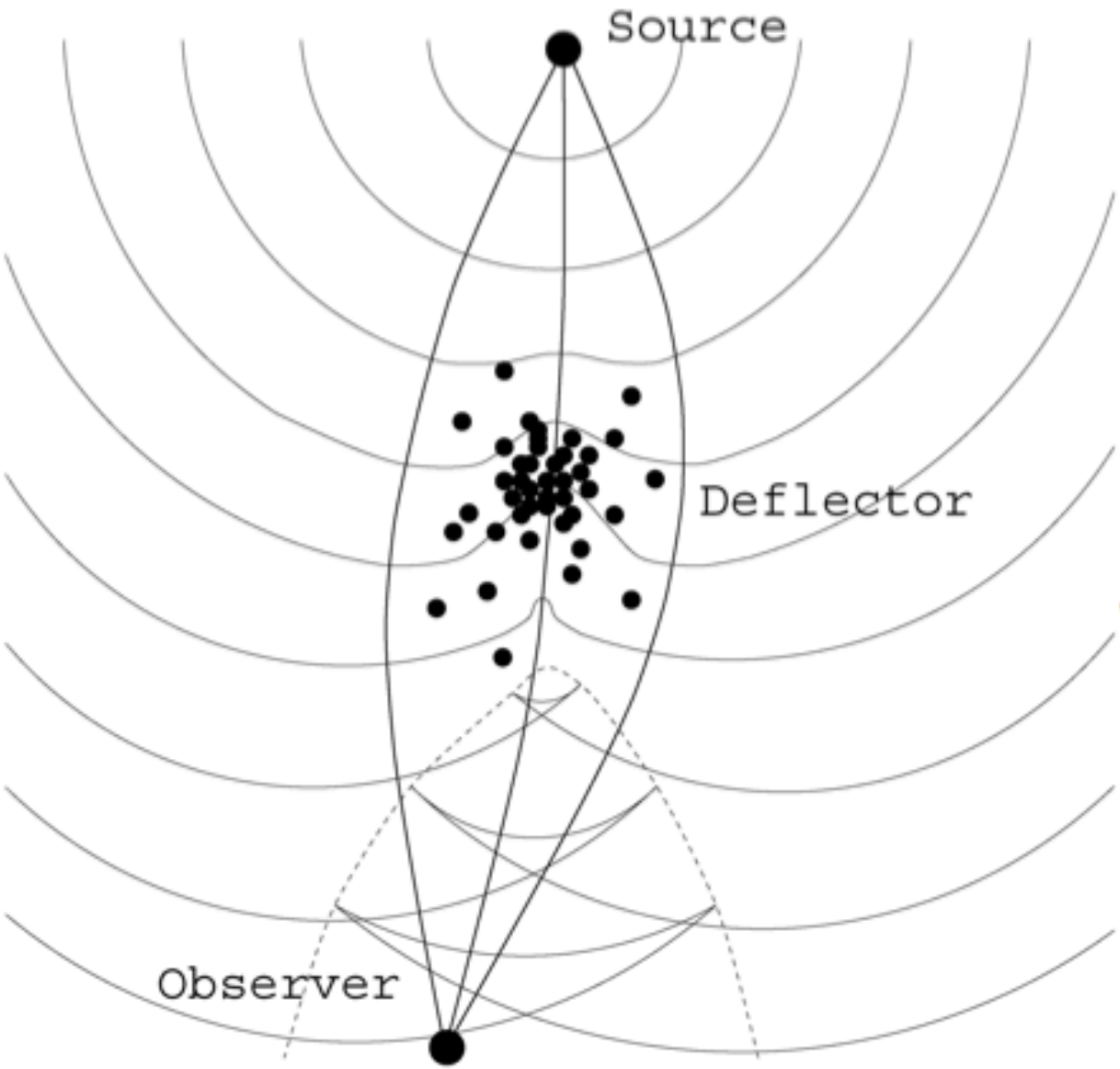




- weak lensing
  - lensing via large-scale structure

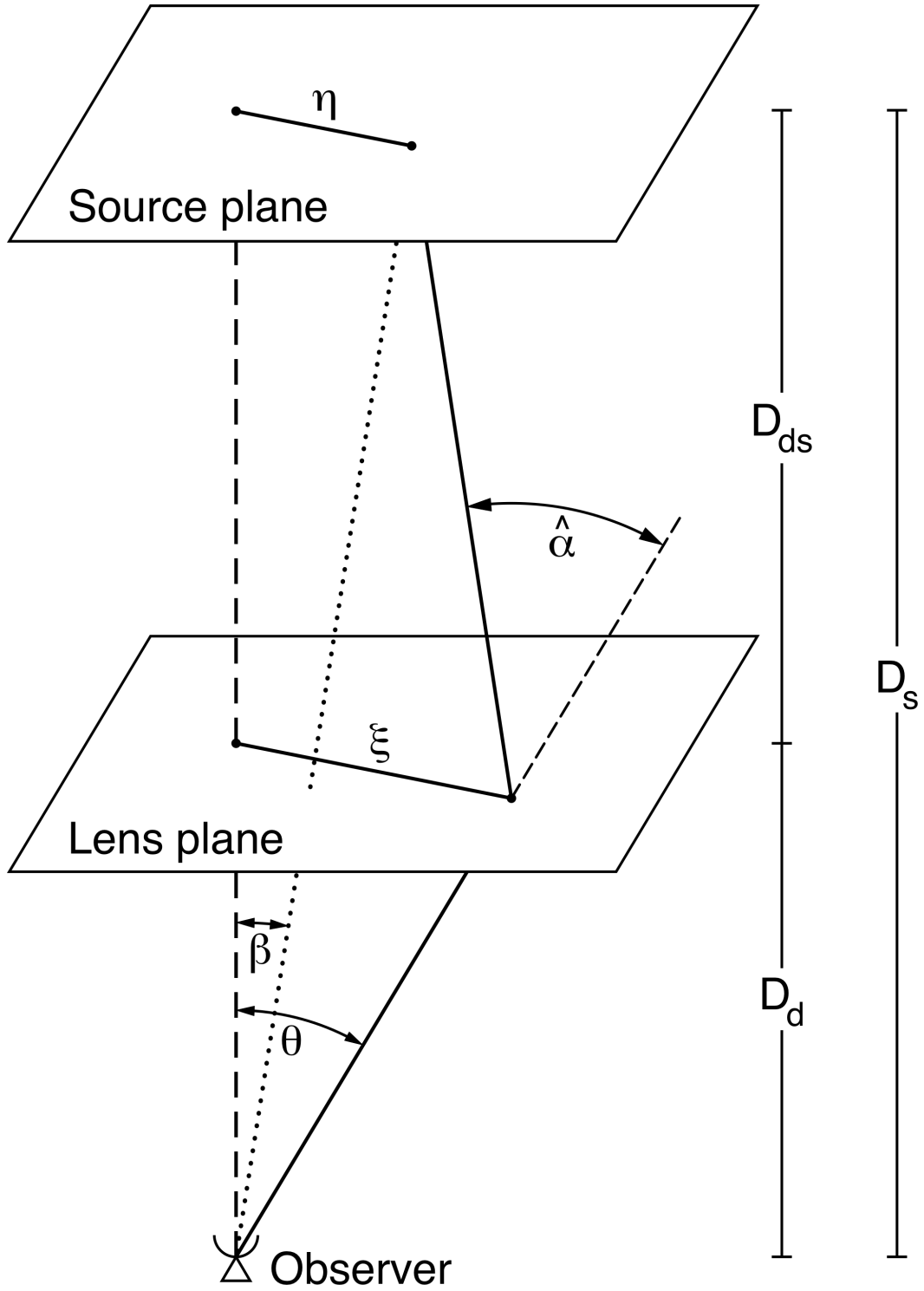
    (→ weak distortion and magnification)

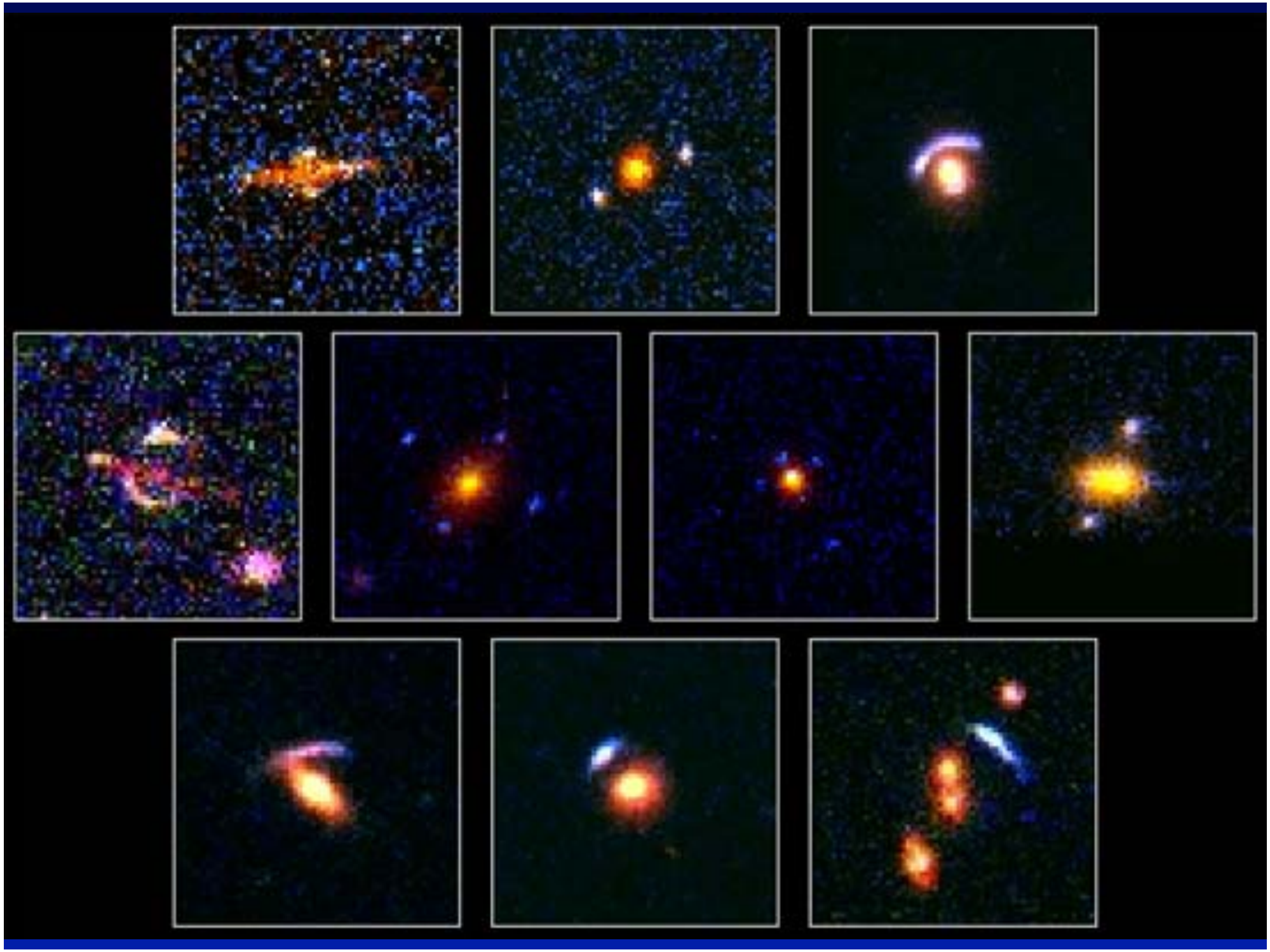
wave picture

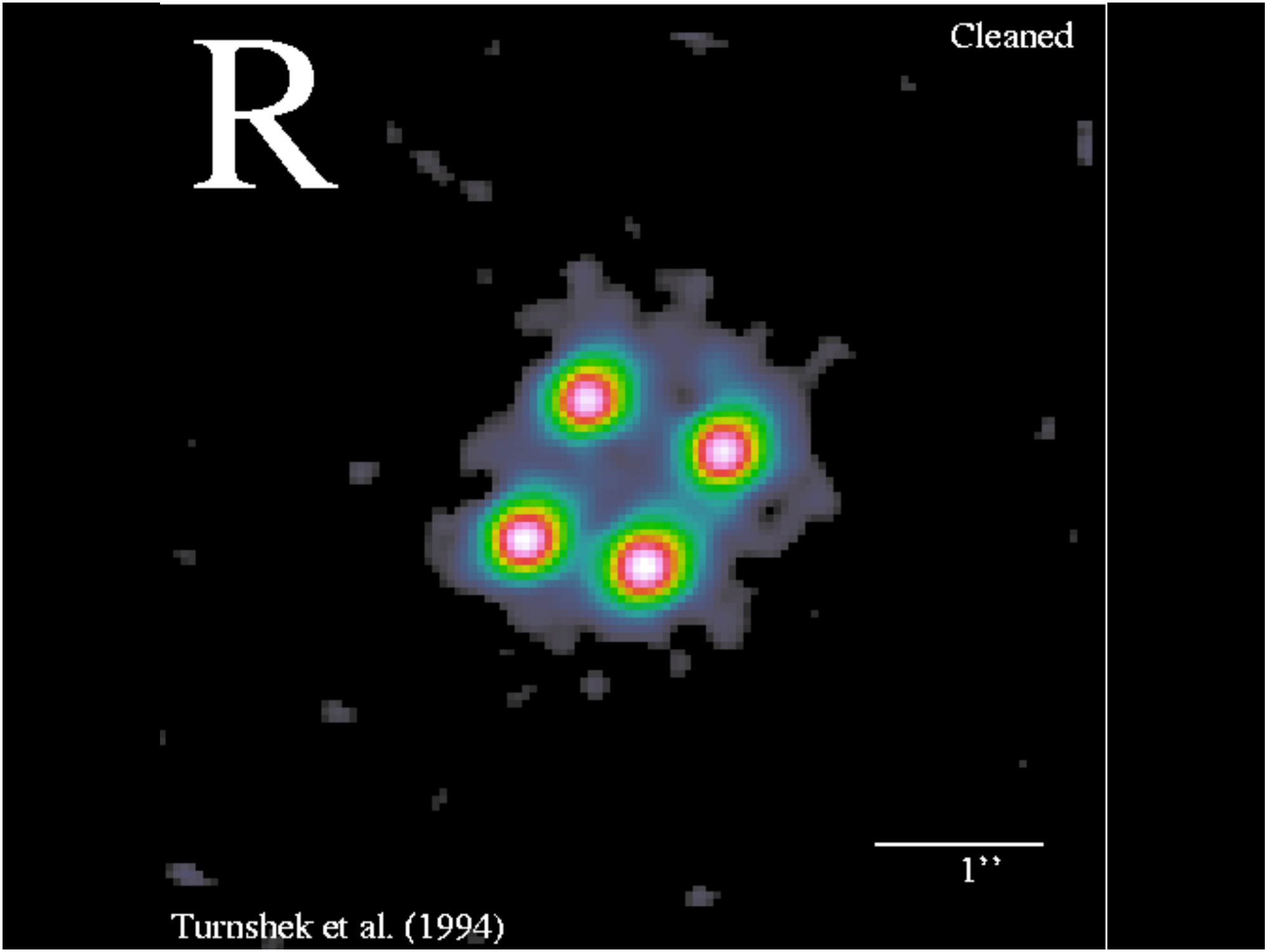


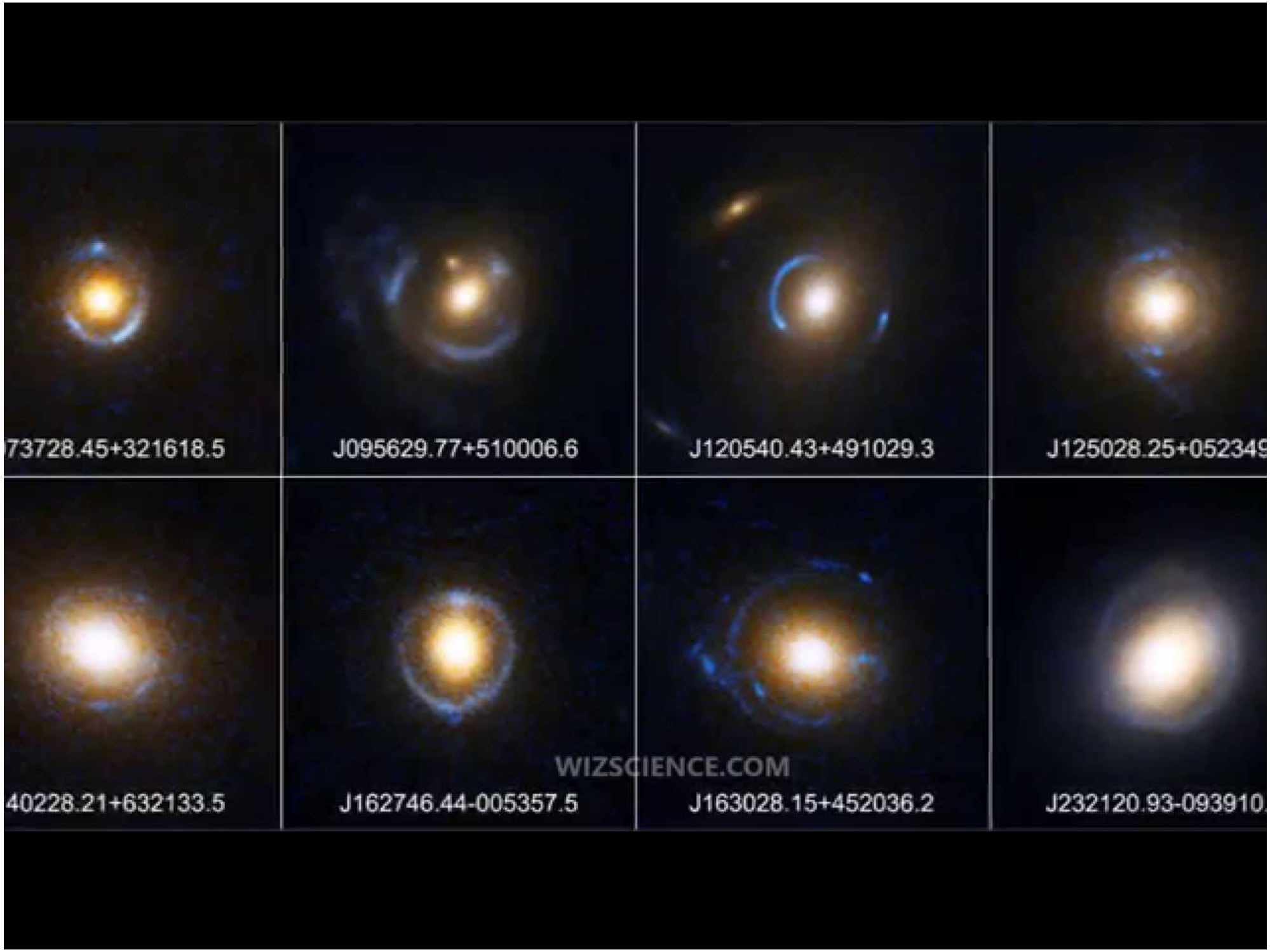
weak lensing regime

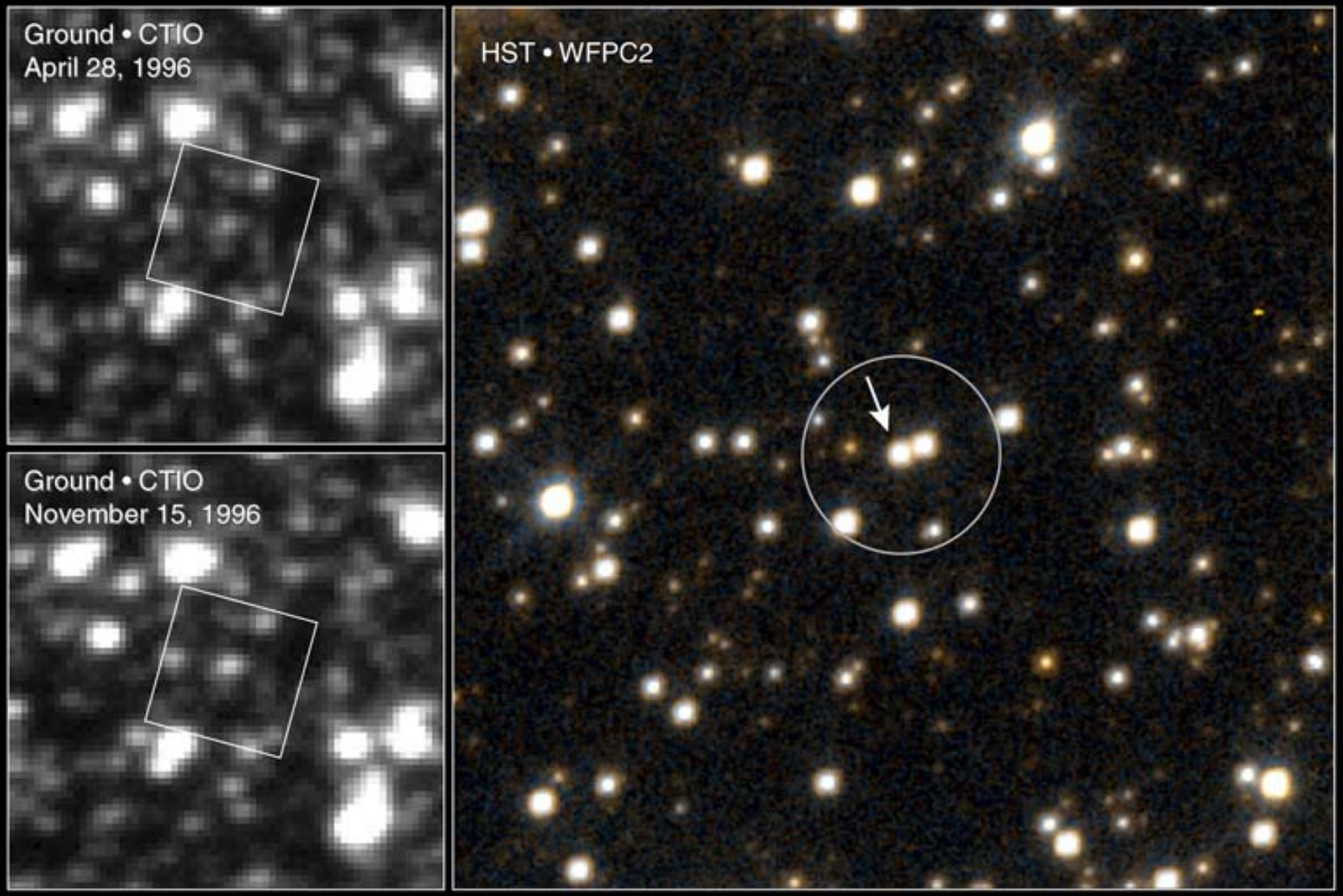
strong/micro lensing regime


η
Source plane
D_{ds}
$\hat{\alpha}$
D_s
ξ
Lens plane
β
θ
D_d
Observer




R
Cleaned
1''
Turnshek et al. (1994)

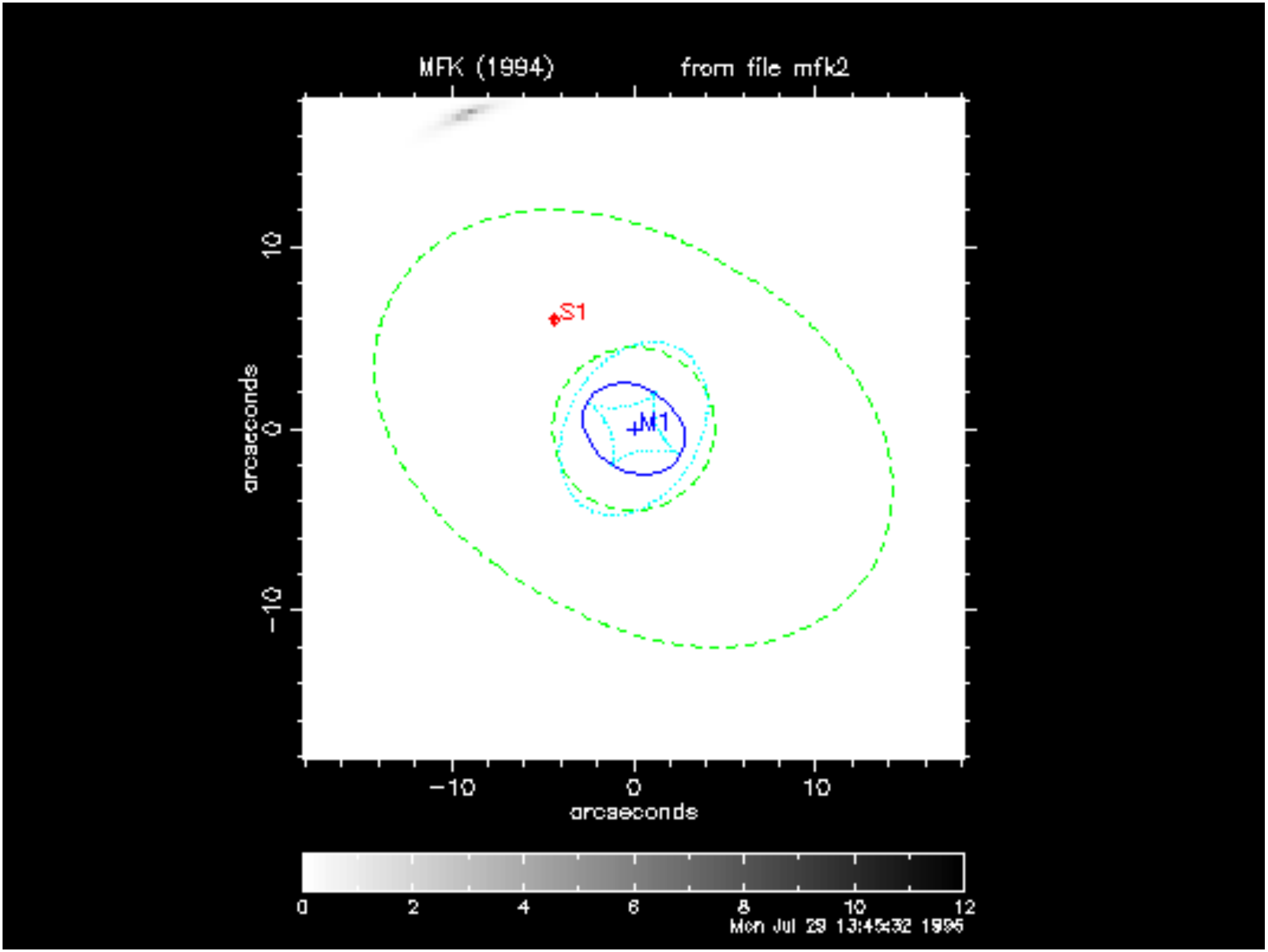

73728.45+321618.5
J095629.77+510006.6
J120540.43+491029.3
J125028.25+052349
40228.21+632133.5
J162746.44-005357.5
J163028.15+452036.2
J232120.93-093910.
WIZSCIENCE.COM



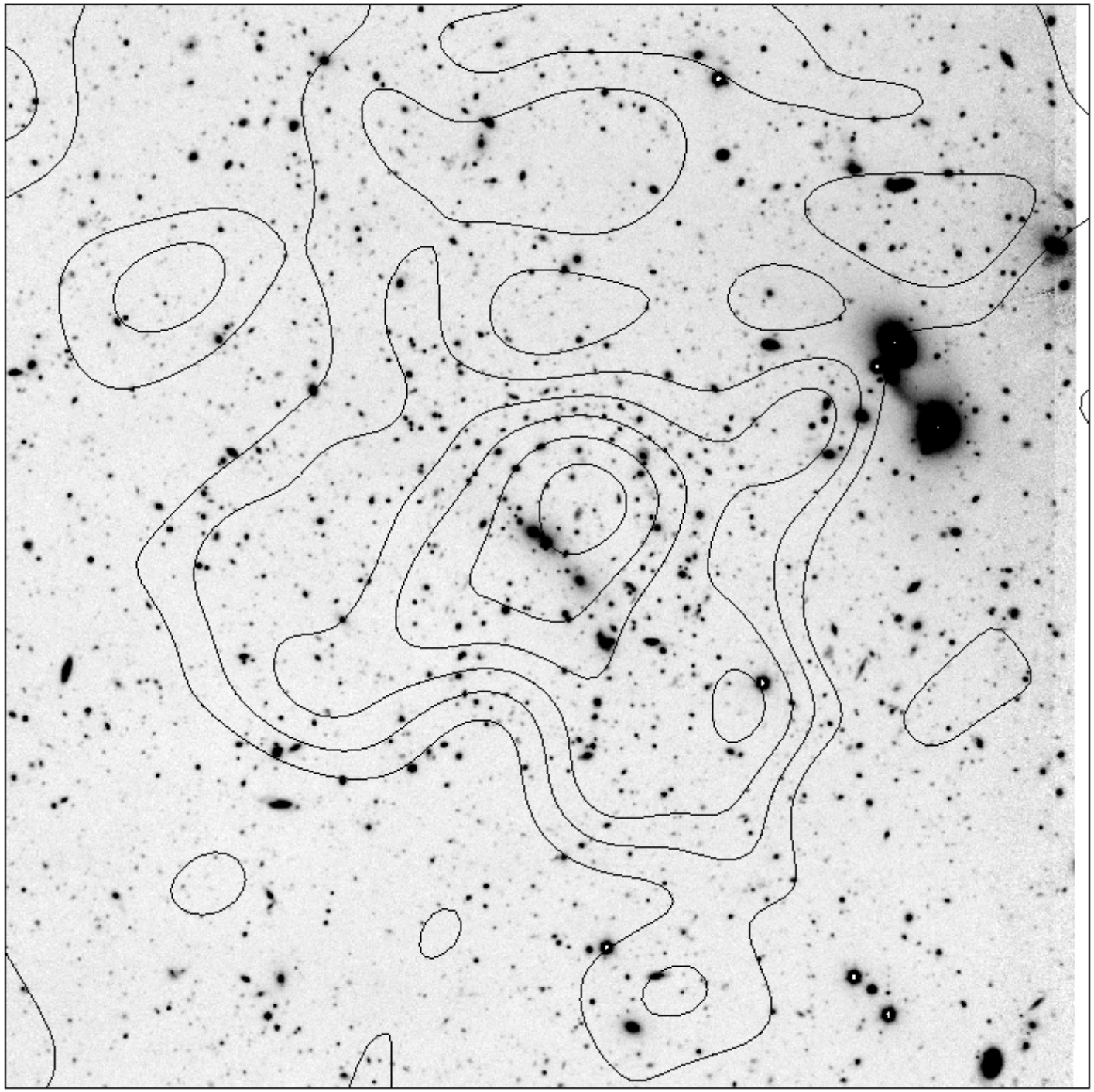
Microlens Event MACHO-96-BLG-5

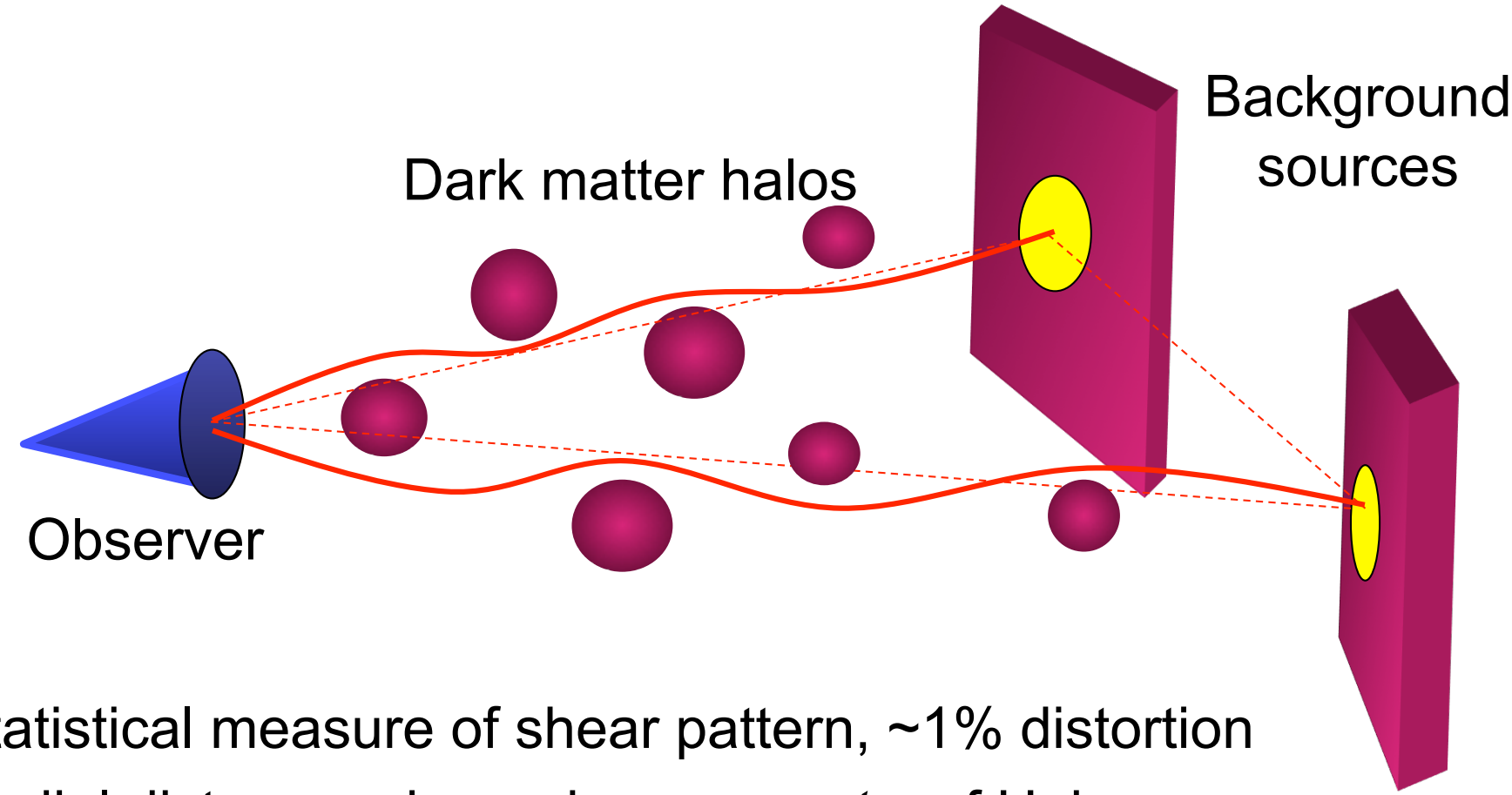
Hubble Space Telescope • WFPC2

NASA and D. Bennett (University of Notre Dame) • STScI-PRC00-03

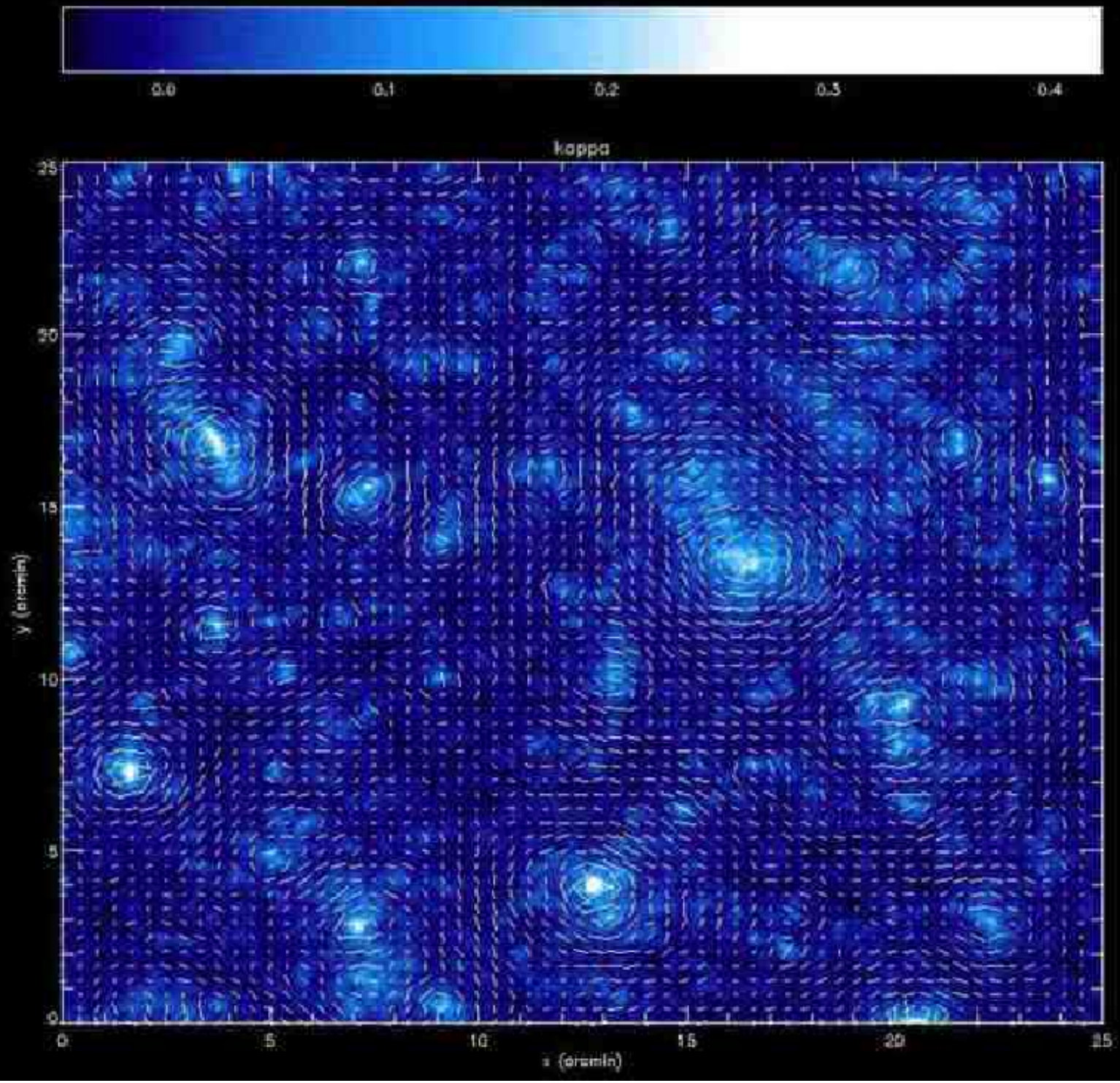

MFK (1994)
from file mfk2
S1
M1
10
0
−10
arcseconds
−10
0
10
arcseconds
0
2
4
6
8
10
12
Mon Jul 29 13:45:32 1996

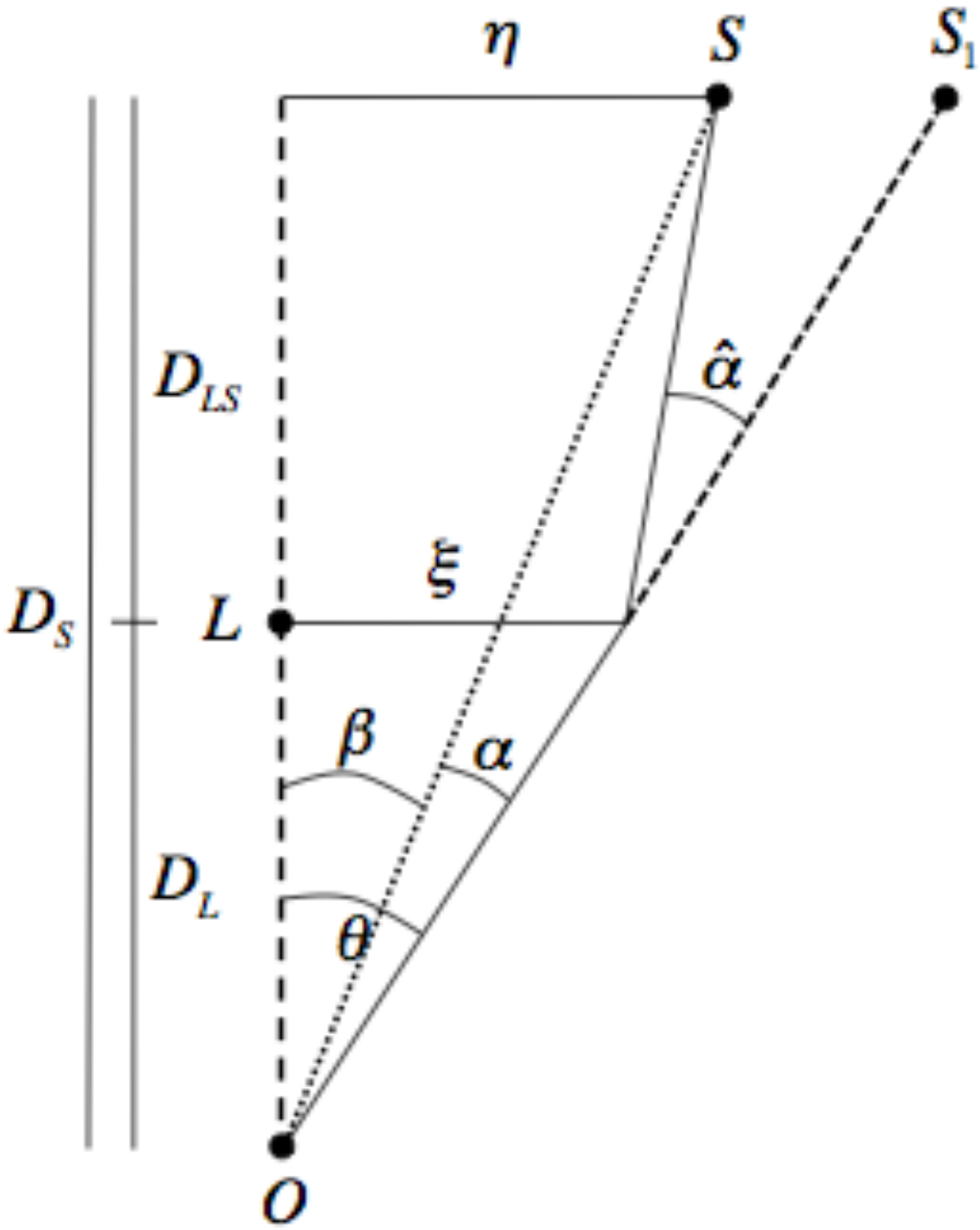
MS0016+16



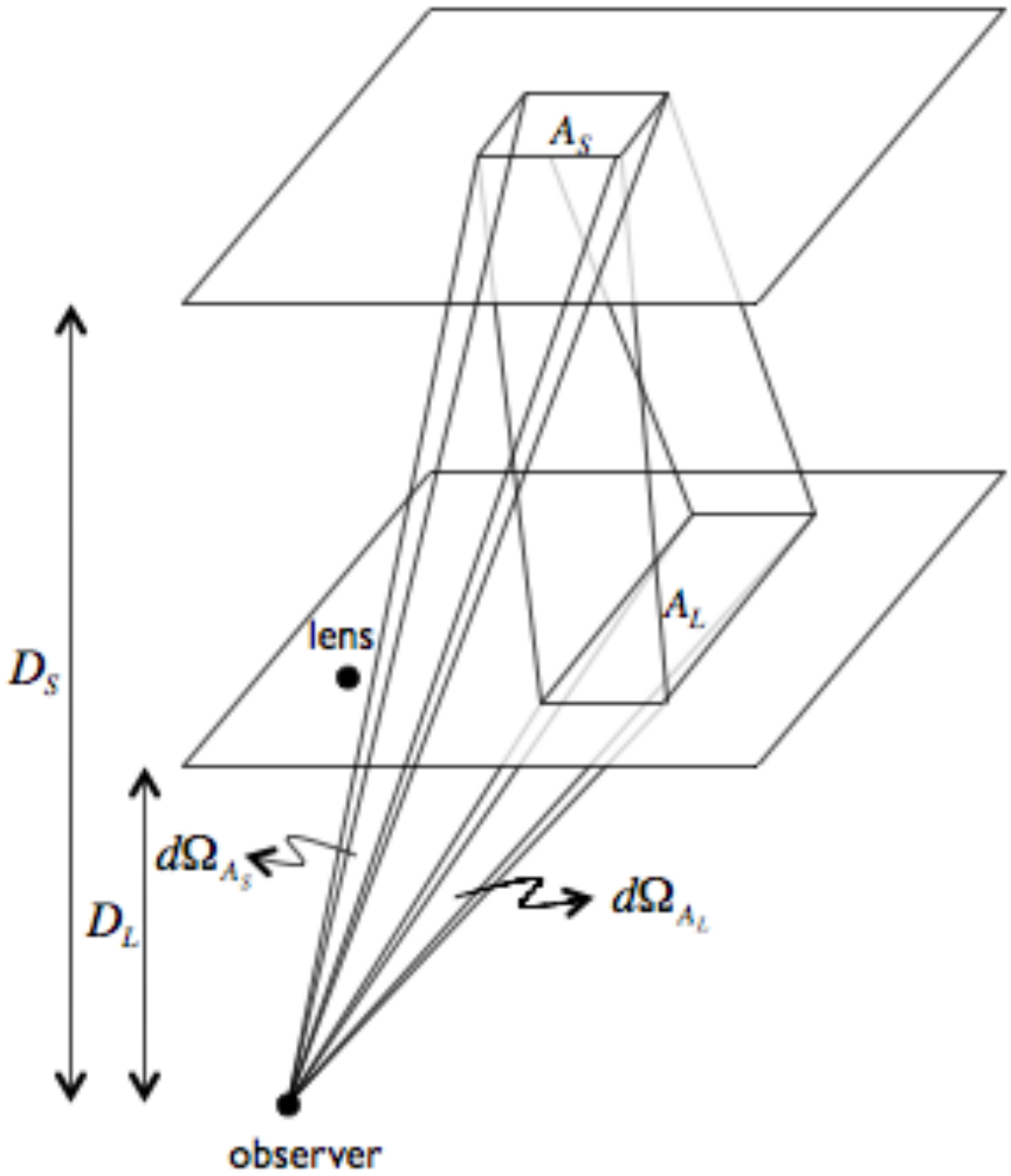


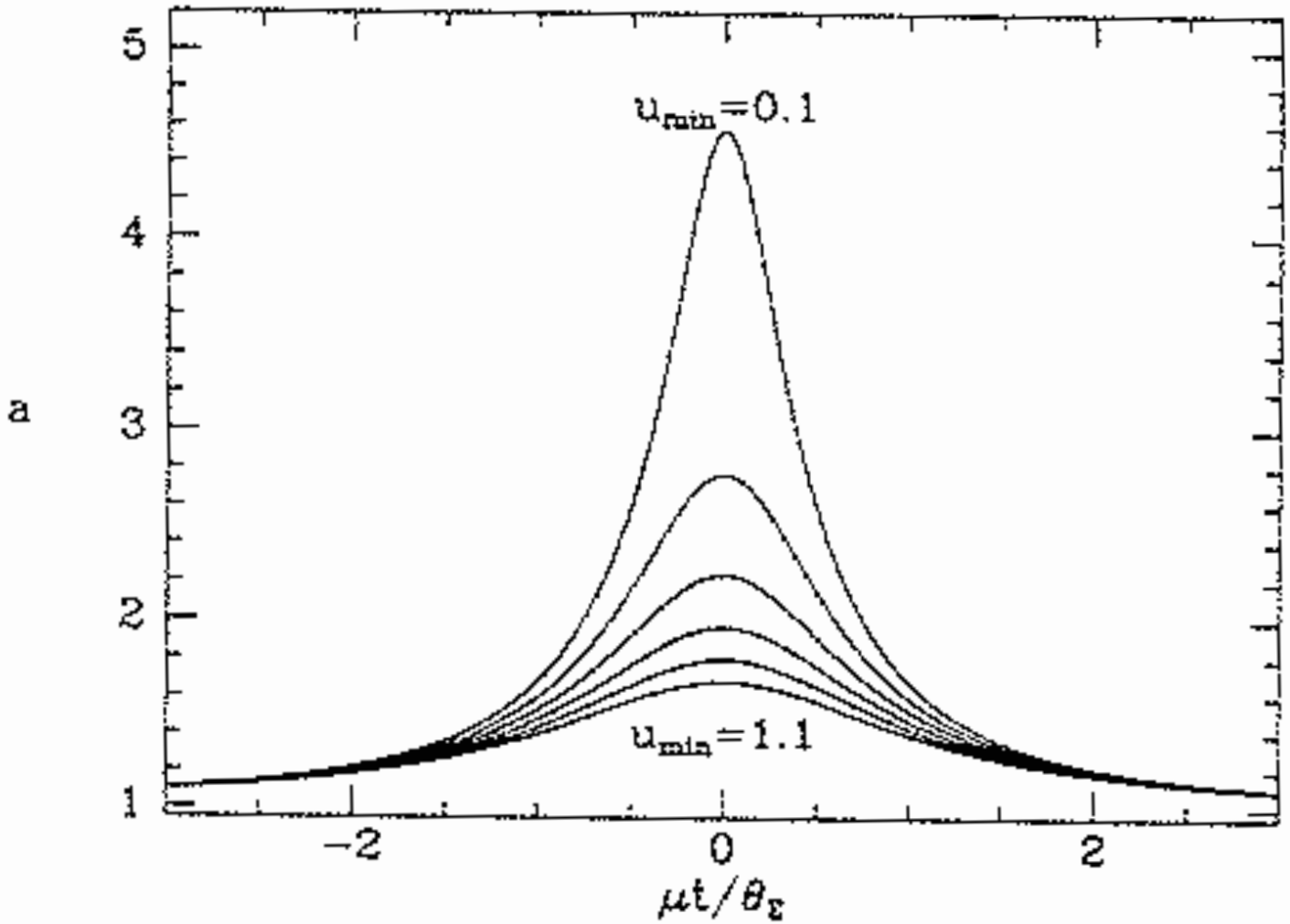
- Statistical measure of shear pattern, ~1% distortion
- Radial distances depend on *geometry* of Universe
- Foreground mass distribution depends on *growth* of structure


0.0
0.1
0.2
0.3
0.4
kappa
25
20
15
10
5
0
y (arcmin)
0
5
10
15
20
25
x (arcmin)

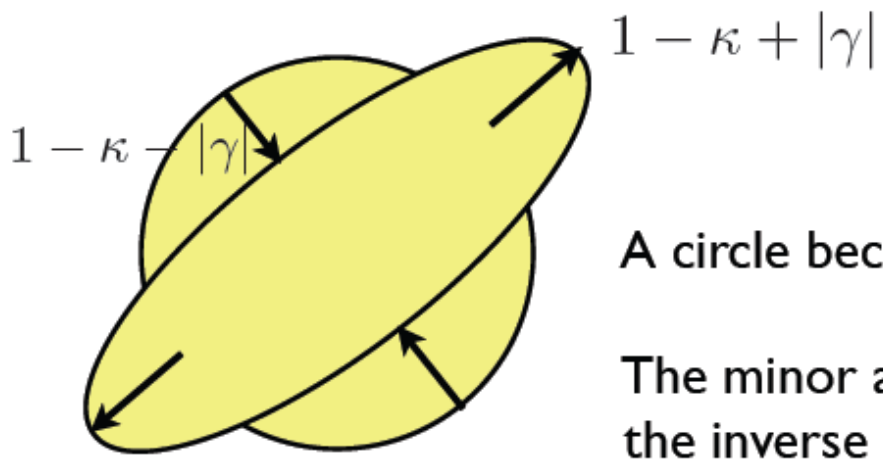

η
S
S1
DLS
α̂
ξ
DS
L
β
α
DL
θ
O

lensing preserves surface brightness




a
5
4
3
2
1
$u_{min}=0.1$
$u_{min}=1.1$
−2
0
2
$\mu t/\theta_E$

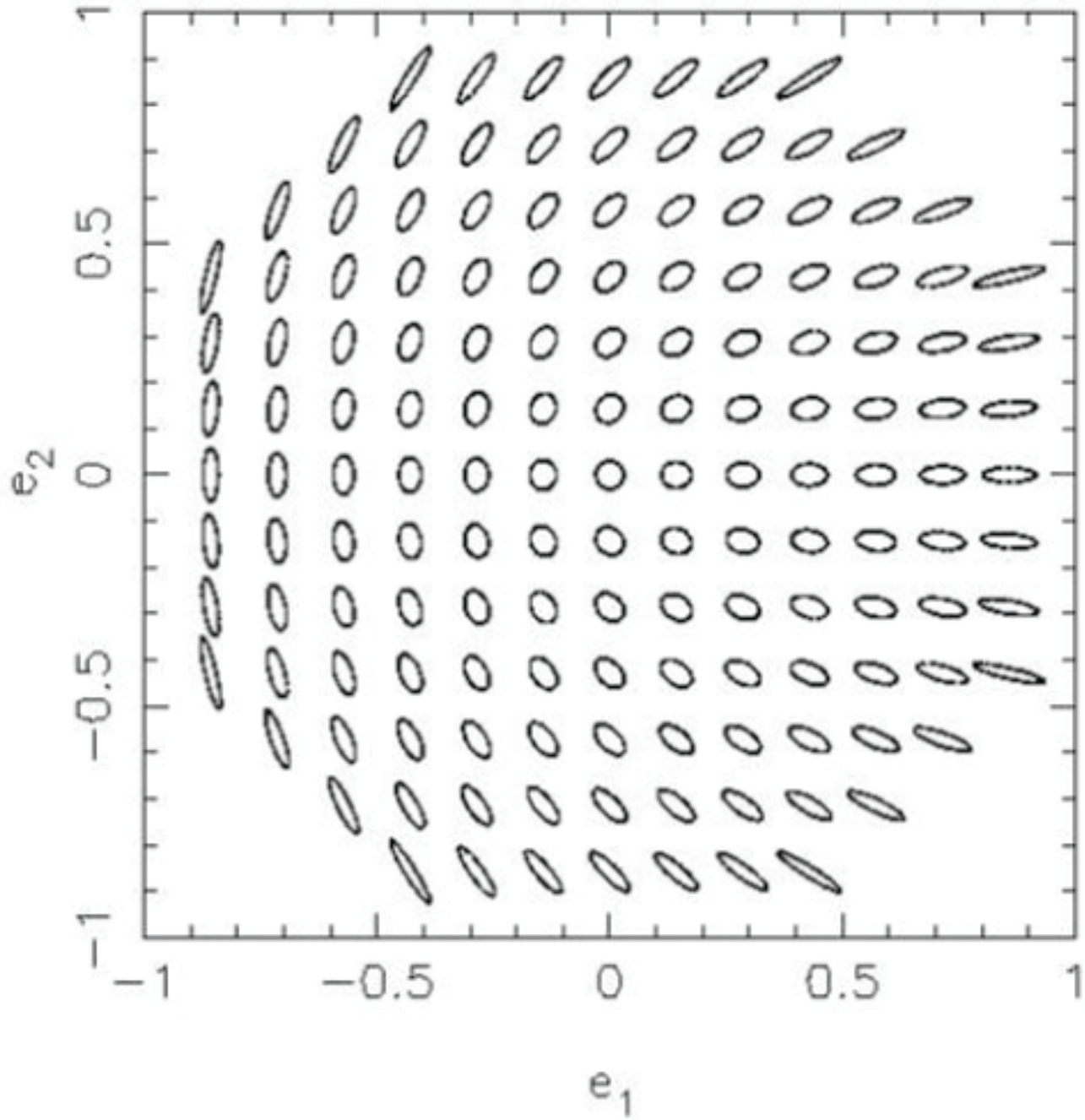
$$I(\vec{\theta}) = I^{(s)}\left[\vec{\beta}_0 + \mathcal{A}(\vec{\theta}_0)\cdot(\vec{\theta} - \vec{\theta}_0)\right]$$

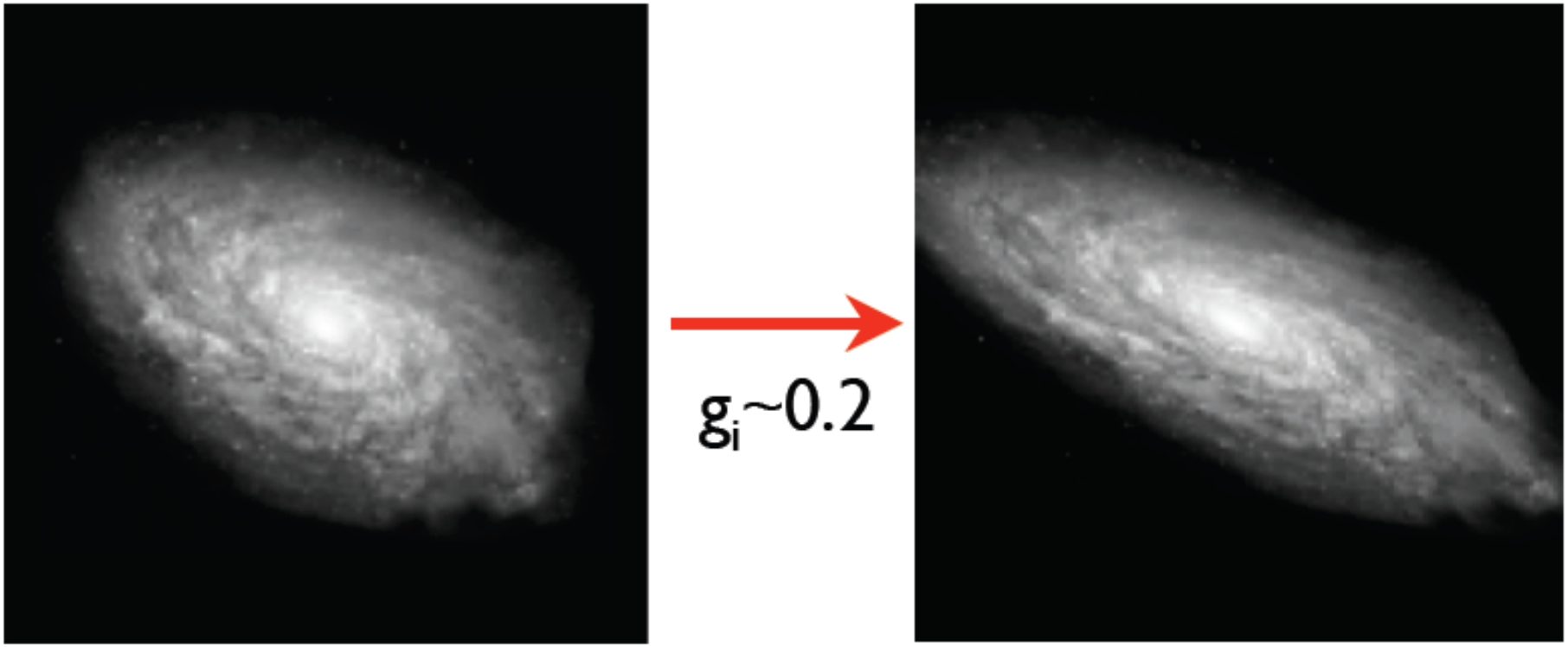


A circle becomes an ellipse.

The minor and major axes given by the inverse of the eigenvalues of A.

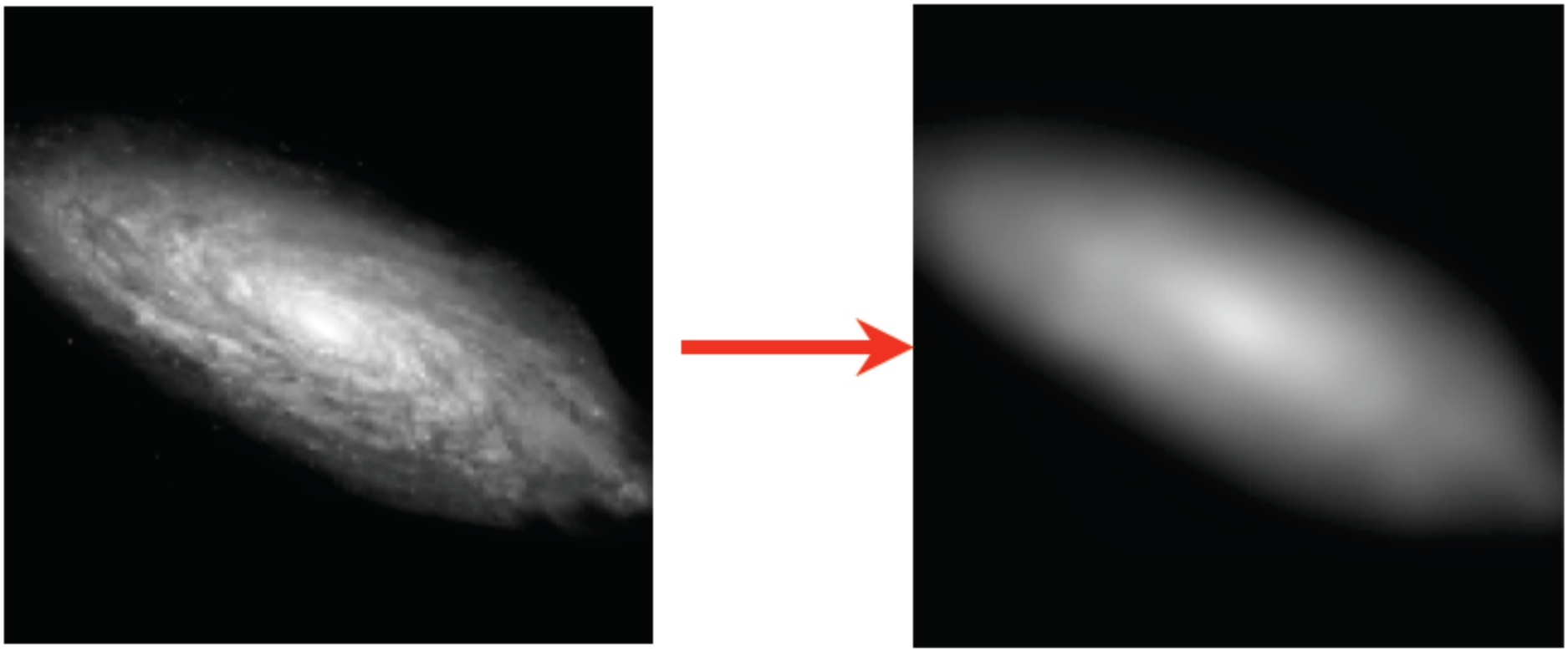
The orientation is given by the eigenvectors of A


1
0.5
0
−0.5
−1
e2
−1
−0.5
0
0.5
1
e1



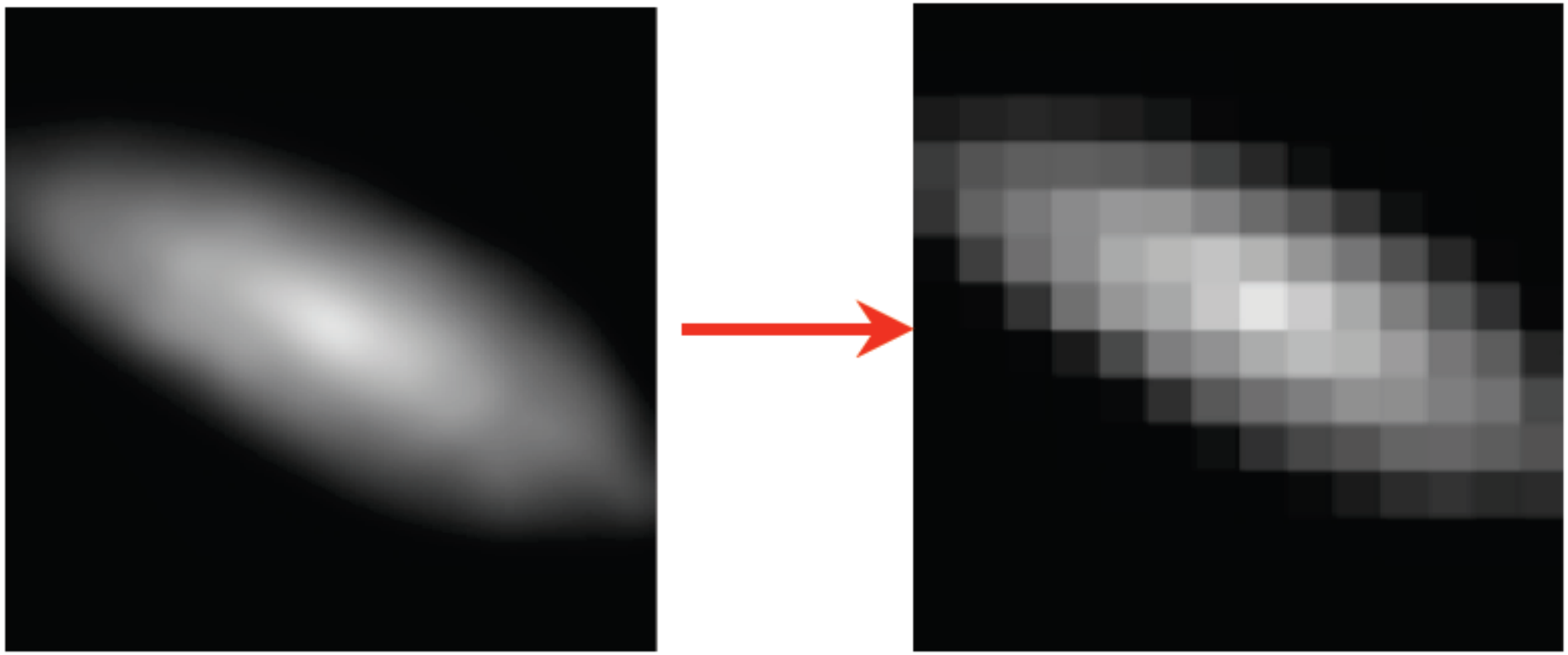
$$\begin{pmatrix} x_u \\ y_u \end{pmatrix} = \begin{pmatrix} 1-g_1 & -g_2 \\ -g_2 & 1+g_1 \end{pmatrix} \begin{pmatrix} x_l \\ y_l \end{pmatrix}$$

Real data:
g_i~0.03



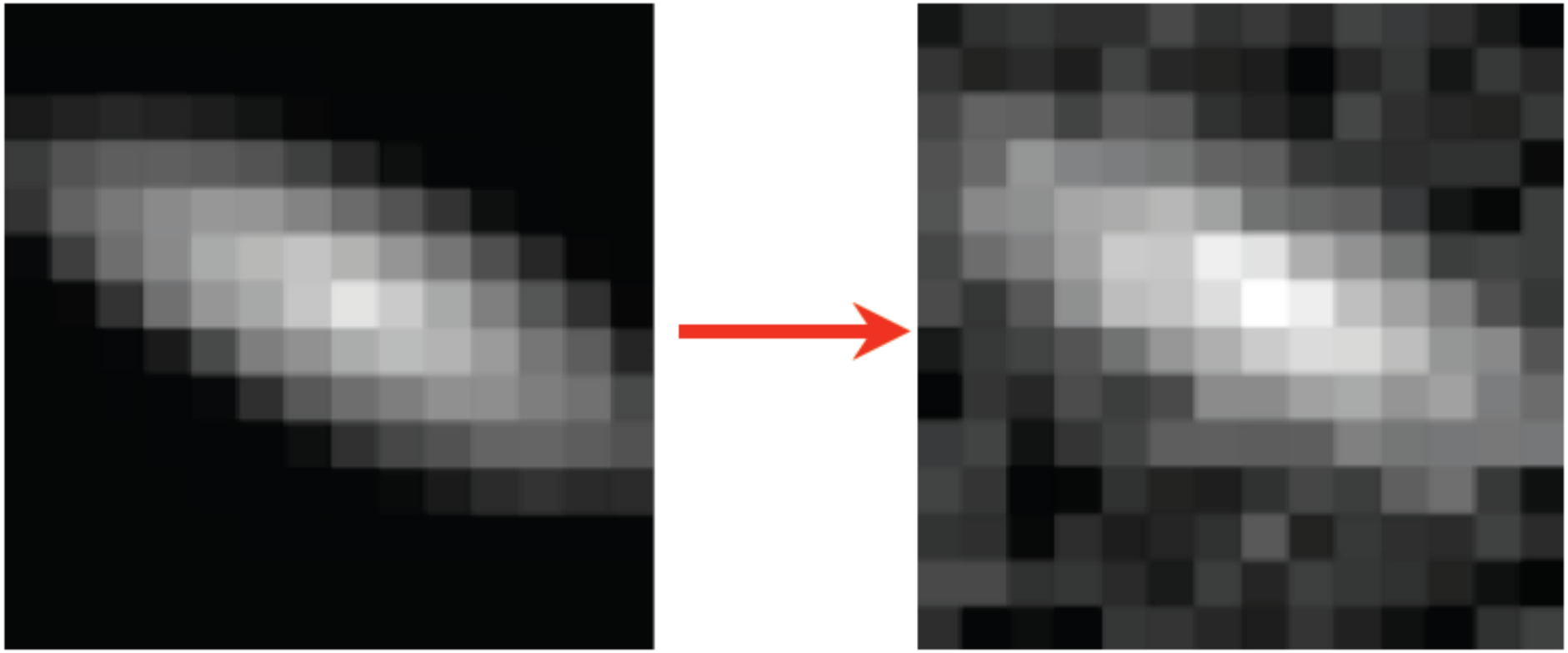
Atmospheric Seeing and telescope PSF

Real data: seeing disk ~ Galaxy size



Sum light in each square

Real data: Pixel size ~ seeing size /3



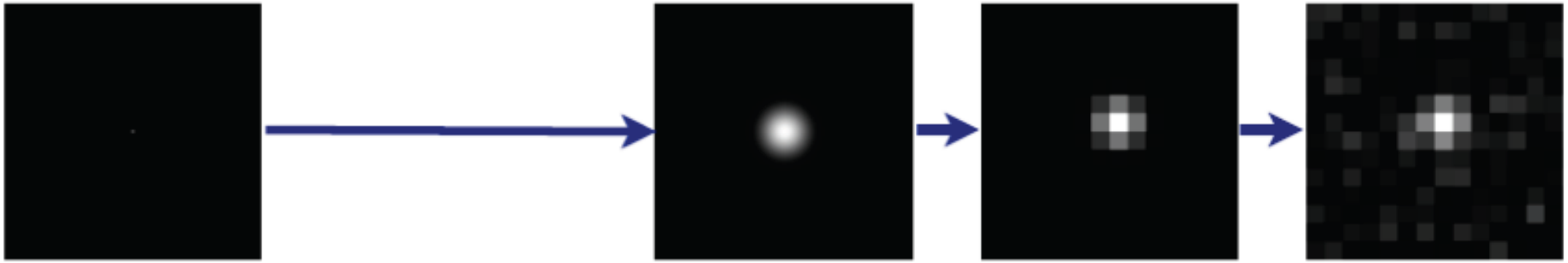
Mostly Poisson. Some Gaussian and bad pixels.
Uncertainty on total light ~ 5 per cent

The Forward Process.

Galaxies: Intrinsic galaxy shapes to measured image:

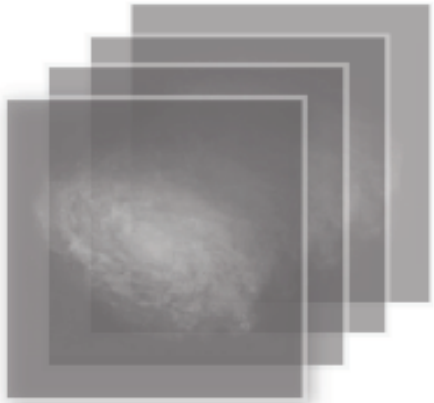
Intrinsic galaxy (shape unknown) → Gravitational lensing causes a ***shear (g)*** → Atmosphere and telescope cause a convolution → Detectors measure a pixelated image → Image also contains noise

Stars: Point sources to star images:



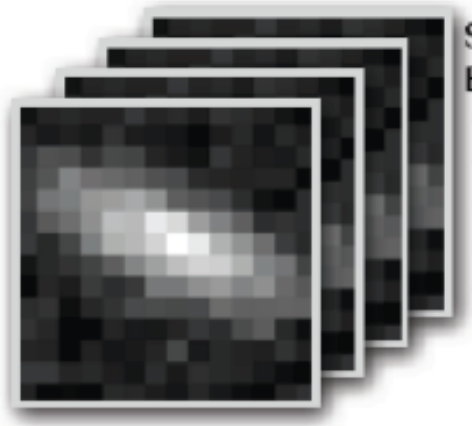
Intrinsic star (point source) → Atmosphere and telescope cause a convolution → Detectors measure a pixelated image → Image also contains noise

The Inverse Problem:
Measured images to *shear*

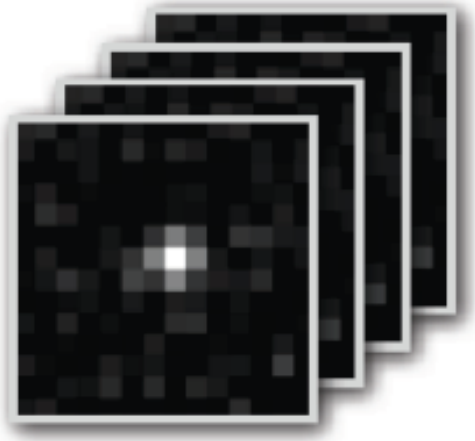


Intrinsic galaxy shapes can be inferred, but are not used beyond shear estimation

Shear Field

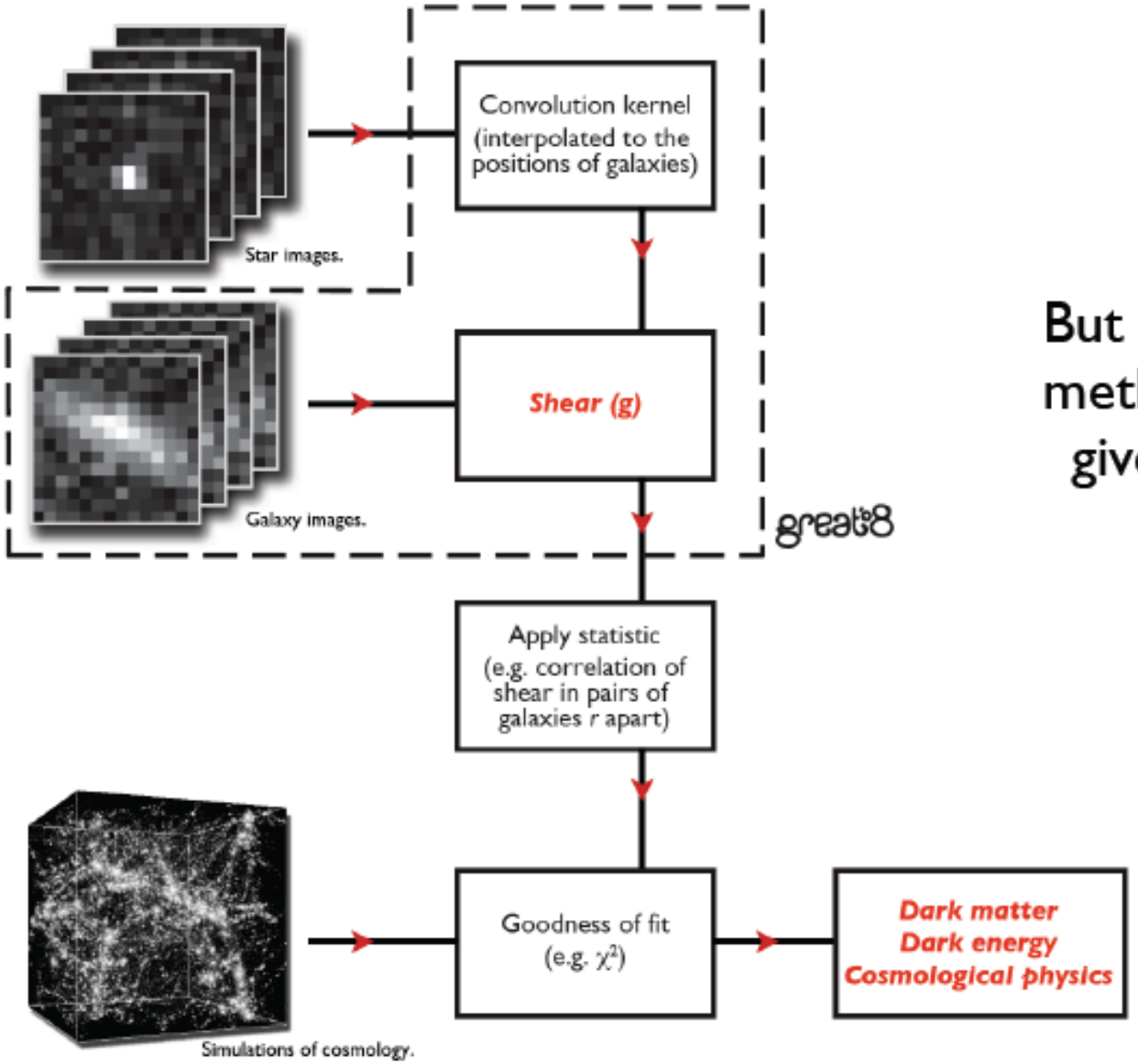


Set of galaxy images.
Each contains:
- noise
- pixelisation
- convolution
- ***shear***
- intrinsic shape



Set of star images.
Each contains:
- noise
- pixelisation
- convolution

A full weak lensing pipeline:
The broader context typical for cosmological measurements



But there are many different methods that were found to give very different results!